The “Sugar Trap” Market Gap Analysis — Helix CPG Partners

# Dashboard Build Guide
# Google Looker Studio

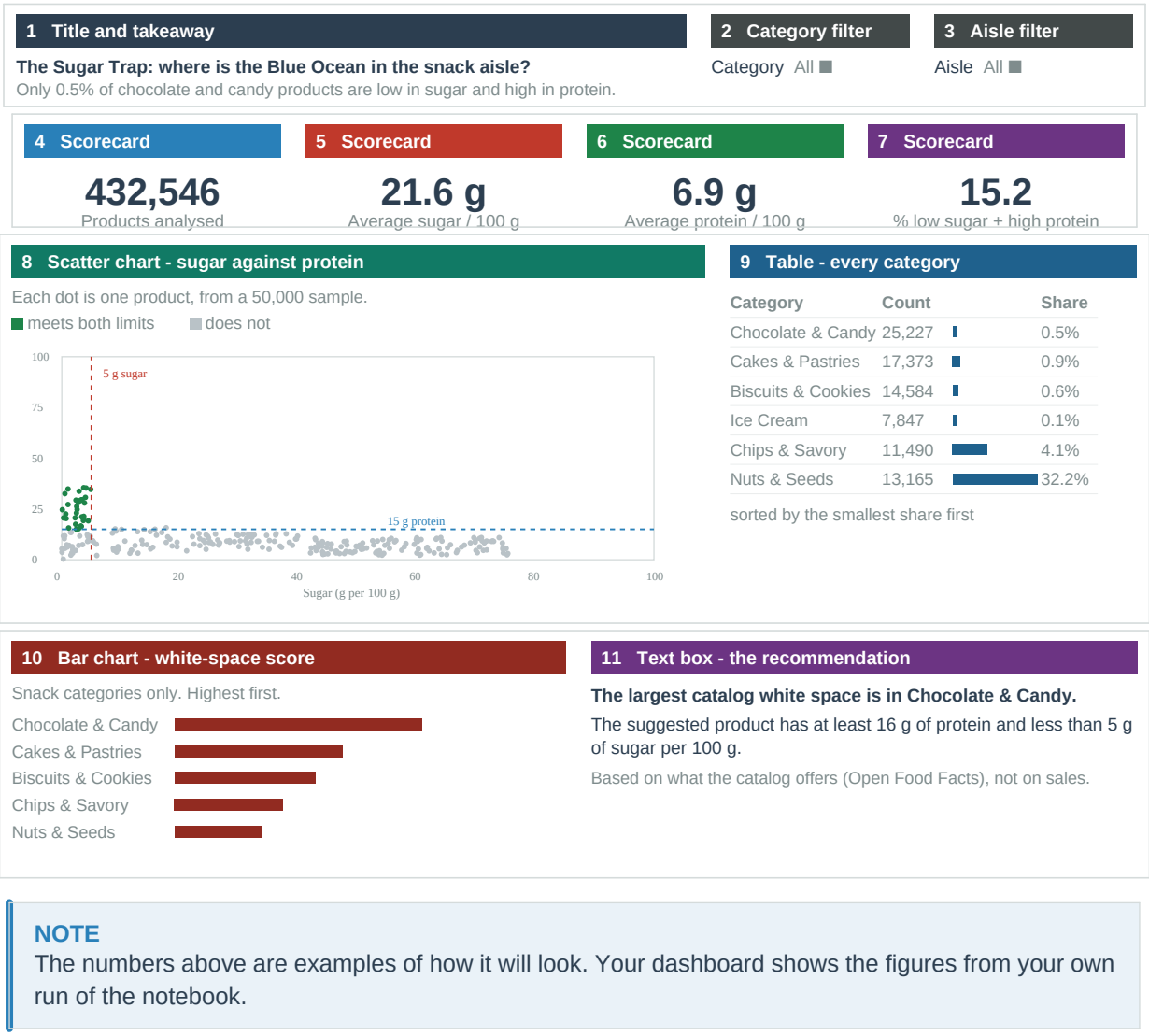
Every click, every field and every formula needed to build the dashboard, element by element, from the three tables the notebook produces.

| | |
|---|---|
| **Tool** | Google Looker Studio (free, sign in with a Google account) |
| **Data** | three CSV files from sugar_trap_dashboard_data.zip |
| **Build time** | about 45 minutes the first time |
| **Result** | one dashboard page with a public link that needs no sign-in |

Companion to sugar_trap_market_gap_analysis.ipynb • Emmanuel ASIFIWE

# 1 How the finished dashboard should look

This is the page you are going to build. Every box is numbered, and section 7 builds them one by one in this order. The page is one screen wide, so nothing needs scrolling.




> **NOTE**
> The numbers above are examples of how it will look. Your dashboard shows the figures from your own run of the notebook.

# 2 Where each element sits

Looker Studio has no box for typing a position, so you place elements by dragging. Blue guide lines appear when an edge lines up with another element. The table below gives the share of the page each element takes, which is all you need to match the design.

| # | Element | Across the page | Down the page |
|---|---|---|---|
| 1 | Title and takeaway (text) | left 60% | top strip |
| 2 | Category filter (control) | next 20% | top strip |
| 3 | Aisle filter (control) | right 20% | top strip |
| 4-7 | Four scorecards | one quarter each | below the title |
| 8 | Scatter chart | left 62% | middle band, the tallest element |
| 9 | Category table | right 38% | middle band, same height as the scatter |
| 10 | White-space bar chart | left 50% | bottom band |
| 11 | Recommendation (text) | right 50% | bottom band |

> **TIP**
> Select several elements with **Shift + click**, then use **Arrange › Align** and **Arrange › Distribute** to line them up exactly. Nudge a selected element one pixel at a time with the arrow keys.

To make the page the right shape, open **Theme and layout** on the toolbar, go to the **Layout** tab and set **Canvas size** to **Custom**, 1200 by 900 pixels. If you prefer a different size, keep the proportions in the table above.

# 3 Before you start

**You need a Google account** (Looker Studio is free) and the file **sugar_trap_dashboard_data.zip**, which the notebook's last cell downloads. Unzip it. It holds three tables.

| File | What is in it | Rows |
|---|---|---|
| `dashboard_products.csv` | one row per product, a random sample, for the scatter plot | 50,000 |
| `category_summary.csv` | one row per category, worked out from every cleaned product | 18 |
| `overall_summary.csv` | one row of headline totals, from every cleaned product | 1 |

> **NOTE**
> Three tables, not one, so that each number is read from the place it was worked out. The totals and the shares come from the two summary tables, which cover **every** cleaned product. Only the scatter plot uses a sample, because a file holding every product is too large to upload.

**Why Looker Studio and not Power BI?** Power BI's web service does not accept a personal Gmail address, and organisations often switch off its public-link feature. Looker Studio works with a normal Google account and gives a link that opens without signing in.

# 4 Step 1 — Put the three tables into Google Sheets

Looker Studio reads Google Sheets reliably, so each file becomes one tab of one spreadsheet.

1. Open a new tab and go to `sheets.new`.
2. Click **Untitled spreadsheet** at the top left and name it **Sugar Trap Data**.
3. Choose **File › Import › Upload**, and drag in `dashboard_products.csv`.
4. In the box that appears set **Import location** to **Replace current sheet**, **Separator type** to **Comma**, and **Convert text to numbers, dates and formulas** to **Yes**. Click **Import data**.
5. Double-click the tab at the bottom and rename it **products**.
6. Click **+** at the bottom left to add a tab. Choose **File › Import › Upload**, pick `category_summary.csv`, and this time set **Import location** to **Insert new sheet**. Rename that tab **categories**.
7. Do the same for `overall_summary.csv` and name the tab **overall**.

> **WATCH OUT**
> Open the **products** tab and look at the `sugars_100g` column. The values must sit against the **right** edge of their cells, which means Sheets read them as numbers. If they sit on the left they came in as text and the scatter plot will stay empty. Import the file again with **Convert text to numbers** switched on.

# 5 Step 2 — Start the report and connect all three tables

1. Go to `lookerstudio.google.com` and sign in.
2. Click **Create** at the top left, then **Report**. Accept the terms if it is your first time.
3. A panel opens asking for data. Choose **Google Sheets**, find **Sugar Trap Data**, select the **products** tab, leave **Use first row as headers** ticked, then click **Add**, and **Add to report** if it asks again.
4. Looker Studio drops a table on the page. Click it once and press **Delete**.
5. Add the second table: **Resource › Manage added data sources › Add a data source**. Choose **Google Sheets**, the same spreadsheet, the **categories** tab, then **Add**.
6. Repeat once more for the **overall** tab, then click **Done**.
7. Name the report: click **Untitled Report** at the top left and type **The Sugar Trap — snack aisle gap**.
8. Set the page shape: click **Theme and layout** on the toolbar, open the **Layout** tab, and set **Canvas size** to **Custom** with width **1200** and height **900**.
9. Still in **Theme and layout**, open the **Theme** tab and pick a light theme such as **Simple**. A white background keeps the chart colours readable.

> **NOTE**
> You should now have three data sources. You can see them under **Resource › Manage added data sources**. Every chart you add will ask which one to use.

# 6 Step 3 — Check the field types and add two formulas

A field's **type** decides how Looker Studio shows it. Open **Resource › Manage added data sources**, click **EDIT** next to a source, and you get a list of its fields with the type on the right. Click a type to change it.

### On the products source

| Field | Type it should be | Why |
|---|---|---|
| `code` | Text | keeps the zero at the front of a barcode |
| `product_name, brands, category` | Text | labels |
| `sugars_100g, proteins_100g` | Number | the two axes of the scatter plot |
| `fiber_100g, fat_100g, salt_100g` | Number | nutrients |
| `energy_kcal_100g, health_score` | Number | numbers |
| `qualifies, is_snack` | Number | they hold 1 or 0, so they can be averaged |

### On the categories source

| Field | Type it should be | Why |
|---|---|---|
| `category` | Text | the label on every chart |
| `products, qualifying` | Number | counts |
| `qualifying_share` | Number › Percent | it holds 0.0047, and Percent shows that as 0.47% |
| `median_sugar, median_protein` | Number | grams per 100 g |
| `white_space_score` | Number | a score between 0 and 1 |
| `white_space_rank, is_snack` | Number | rank, and 1 or 0 |

### On the overall source

| Field | Type it should be | Why |
|---|---|---|
| `top_category, data_last_changed` | Text | words and a date |
| `everything else` | Number | counts and percentages |

> **WATCH OUT**
> `qualifying_share` on the **categories** source is a fraction, so set it to **Percent**. But `qualifying_share_all_%` on the **overall** source is already multiplied by 100, so leave it as a plain **Number**. Setting that one to Percent would show 1520% instead of 15.2%.

## Two formulas to add

These both go on the **products** source. Open **Resource › Manage added data sources › EDIT** on **products**, then click **ADD A FIELD** at the top right.

### Formula 1 — colours the dots on the scatter plot

1. Click **ADD A FIELD**.
2. In **Field Name** type **Meets both limits**.
3. In the **Formula** box type this:

```
CASE WHEN qualifies = 1 THEN "Meets both limits" ELSE "Does not" END
```

4. A green tick saying the formula is valid appears under the box. Click **SAVE**, then **DONE**.

### Formula 2 — lets you show only the snack aisle

1. Click **ADD A FIELD** again.
2. Name it **Aisle**.
3. Type this formula:

```
CASE WHEN is_snack = 1 THEN "Snack aisle" ELSE "Other foods" END
```

4. Click **SAVE**, then **DONE**, then **CLOSE** to come back to the report.

> **WATCH OUT**
> Type the quotation marks yourself rather than copying them from this PDF. A curly quote makes Looker Studio show a red error. Straight quotes look like " and the field name must be spelled exactly as it appears in the field list.

> **NOTE**
> The **categories** and **overall** tables already hold their own percentages, so they need no formulas. Always show their numbers with **Average**, never Sum: each value appears once and adding it up would multiply it.

# 7 Step 4 — Build each element

Each block below builds one numbered box from the design. The pattern is always the same: insert the element, pick its data source at the top of the **Setup** tab on the right, then set the **Dimension** (the label) and the **Metric** (the number).

> **NOTE**
> **How to change how a number is added up.** In the **Setup** tab, hover over the metric and click the small pencil. A box opens with the field **Name** and its **Aggregation**. Choose **Average**, **Sum** or **Count** there. You can also rename the metric in the same box, which changes the label shown on the chart.

| 1 • Title and takeaway | |
|---|---|
| **Insert** | **Insert › Text**, then drag a box across the left 60% of the top strip |
| **Line 1** | type: *The Sugar Trap: where is the Blue Ocean in the snack aisle?* |
| **Line 2** | press Enter and type one sentence of what the reader should notice, taking the figure from the notebook, for example: *Only 0.5% of chocolate and candy products are low in sugar and high in protein.* |
| **Style** | select the first line and set it to about **24 px** and **bold**; set the second line to about **13 px** and grey |

| 2 • Category filter | |
|---|---|
| **Insert** | **Add a control › Drop-down list**, then place it in the top strip |
| **Data source** | **products** |
| **Control field** | `category` |
| **Metric** | leave it as **Record Count**; it only decides the order of the list |
| **Style** | in the **Style** tab tick **Search box** so the list can be typed into |
| **What it does** | changes every chart built on the **products** source, which is the scatter plot and two of the scorecards |

| 3 • Aisle filter | |
|---|---|
| **Insert** | **Add a control › Drop-down list**, placed beside the first one |
| **Data source** | **products** |
| **Control field** | `Aisle`, the formula you added in step 3 |
| **Use** | choose *Snack aisle* to show only the seven snack categories |

## The four scorecards (4 to 7)

A scorecard shows a single number. Insert each one with **Insert › Scorecard**, then set its data source and its one metric. Give each a short title with **Style › Show title**, or place a small text box underneath.

| 4 • Scorecard — products analysed | |
|---|---|
| **Data source** | **overall** |
| **Metric** | `products_after_cleaning`, aggregation **Average** |
| **Why Average** | the table has one row, so Average returns that row's value unchanged |
| **Style** | **Style › Decimal precision 0**, and tick **Compact numbers** if you prefer 433K to 432,546 |
| **Label** | Products analysed |

| 5 • Scorecard — average sugar | |
|---|---|
| **Data source** | **products** |
| **Metric** | `sugars_100g`, aggregation **Average** |
| **Style** | **Decimal precision 1** |
| **Label** | Average sugar per 100 g |
| **Note** | this one reads the 50,000 product sample, so it is very close to, but not exactly, the figure for every product |

| 6 • Scorecard — average protein | |
|---|---|
| **Data source** | **products** |
| **Metric** | `proteins_100g`, aggregation **Average** |
| **Style** | **Decimal precision 1** |
| **Label** | Average protein per 100 g |

| 7 • Scorecard — share that meets both limits | |
|---|---|
| **Data source** | **overall** |
| **Metric** | `qualifying_share_all_%`, aggregation **Average** |
| **Style** | **Decimal precision 1** |
| **Label** | % low sugar + high protein |
| **Why this source** | it covers every cleaned product, not the sample, so the headline share is exact. Leave the field as a plain Number and put the % sign in the label. |

| 8 • Scatter chart — sugar against protein (the main chart) | |
|---|---|
| **Insert** | **Insert › Scatter chart**, then drag it to fill the left 62% of the middle band |
| **Data source** | **products** |
| **Dimension** | `code`. This is the barcode, so each dot is one product. Do not use the category here, or you will get one dot per category. |
| **Metric X** | `sugars_100g`, aggregation **Average** |
| **Metric Y** | `proteins_100g`, aggregation **Average** |
| **Bubble size** | leave empty |
| **Colour** | if the Setup tab shows a **Breakdown Dimension** box, put `Meets both limits` in it, then set the two colours in the **Style** tab: green for *Meets both limits*, light grey for *Does not*. If there is no such box, leave every dot one colour; the two lines below still mark the corner. |
| **Line 1** | **Style › Reference lines › Add a reference line**. Set **Axis** to **X**, **Type** to **Constant**, **Value** to **5**. Label it *5 g sugar* and colour it red. |
| **Line 2** | Add a second reference line. **Axis Y**, **Constant**, **Value 15**, label *15 g protein*, coloured blue. |
| **Axis names** | **Style › X axis** and **Y axis**: name them *Sugar (g per 100 g)* and *Protein (g per 100 g)* |
| **Axis range** | set both axes to a maximum of **100** so the shape of the cloud stays the same when a filter is used |
| **Chart title** | **Style › Chart header**, or a text box above the chart: *Sugar against protein — a sample of 50,000 products* |
| **If it is slow** | Looker Studio limits how many dots it draws at once. Use the category filter to look at one category at a time. The corner you are pointing at stays visible either way. |

| 9 • Table — every category with its size | |
|---|---|
| **Insert** | **Insert › Table**, placed in the right 38% of the middle band |
| **Data source** | **categories** |
| **Dimension** | `category` |
| **Metric 1** | `products`, aggregation **Average**. Click the pencil and rename it **Products**. |
| **Metric 2** | `qualifying_share`, aggregation **Average**. Rename it **Qualifying**. |
| **Show as bars** | **Style** tab, find **Metric 2**, and change **Number** to **Bar**. The share then appears as a small bar in each row. |
| **Sort** | **Setup › Sort**: choose `qualifying_share` and **Ascending**, so the emptiest categories are at the top |
| **Rows** | **Style › Rows per page**: set 20 so all 18 categories fit without paging |
| **Why both** | a share means nothing on its own. Showing the size next to it stops a category with nine products looking like a finding. |

| 10 • Bar chart — the white-space score | |
|---|---|
| **Insert** | **Insert › Bar chart**, placed in the left half of the bottom band |
| **Data source** | **categories** |
| **Dimension** | `category` |
| **Metric** | `white_space_score`, aggregation **Average** |
| **Filter** | **Setup › Filter › Add a filter › CREATE A FILTER**. Name it **Snack aisle only**, then set: **Include**, field `is_snack`, condition **Equal to (=)**, value **1**. Click **SAVE**. Only snack categories have a score. |
| **Sort** | **Setup › Sort**: `white_space_score`, **Descending** |
| **Bars sideways** | in the **Style** tab tick **Horizontal** if the option is there, so the category names are easy to read |
| **Title** | a text box above it: *Which gap to work on first*, and underneath in small grey type: *Based on what the catalog offers, not on sales* |
| **What to say** | the top bar is the biggest gap: a large category with almost no low-sugar, high-protein product. It is a candidate to develop, not proof that it will sell. |

| 11 • Text box — the recommendation | |
|---|---|
| **Insert** | **Insert › Text**, placed in the right half of the bottom band |
| **What to paste** | the first paragraph of `recommendation.txt`, which the notebook saves in the outputs folder. It names the category and the grams of protein and sugar. |
| **Second line** | *Based on what the catalog offers (Open Food Facts), not on sales.* |
| **Style** | about **15 px**, and give the box a light background with **Style › Background and border** so it reads as the conclusion |

# 8 Step 5 — Tidy up, then publish

- **Line things up.** Select a row of elements with Shift + click, then **Arrange › Align › Top** and **Arrange › Distribute › Horizontally**.
- **Keep the colours consistent.** Green always means “meets both limits”, grey always means it does not.
- **Check it as a reader.** Click **View** at the top right to see the dashboard without the editing tools. Click **Edit** to come back.
- **Stay on one page.** A single tidy page reads far better in a two minute walkthrough.

> **WATCH OUT**
> This next step is the one people get wrong, and the submission form does not accept a broken link as an excuse. Publish it, then test it yourself.

1. Click **Share** at the top right.
2. In **Manage access**, change **Restricted** to **Anyone with the link**, and leave the role as **Viewer**.
3. Go to **Resource › Manage added data sources**. For each of the three sources click **EDIT**, then **Data credentials**, choose **Owner's credentials**, and click **Update**. This lets a visitor see the data without needing their own access to your spreadsheet.
4. Click **Share › Copy link**. That link is what you submit.

> **NOTE**
> **Test it the way the markers will.** Open a new Incognito or Private window with **Ctrl + Shift + N**, paste the link, and check that the charts appear and nothing asks you to sign in. If it asks for access, go back and check both the sharing setting and the data credentials.

# 9 Before you submit

- [ ] The link opens in an Incognito window with no sign-in.
- [ ] The scatter plot is there, its title says it is a sample, and the category filter changes it.
- [ ] The two reference lines at 5 g sugar and 15 g protein are drawn and labelled.
- [ ] The category table shows each category's size next to its qualifying share.
- [ ] The white-space bar chart is there and is filtered to the snack aisle.
- [ ] The recommendation box holds the sentence from the notebook and the line about catalog data.
- [ ] The four scorecards show sensible numbers, not blanks or zeros.
- [ ] The link is in the README under Project Links, and in the submission form.

# 10 If something goes wrong

| What you see | What to do |
|---|---|
| The scatter plot shows one dot | The dimension is a category. Change it to `code`, and set both X and Y to **Average**. |
| The scatter plot is empty | Sugar and protein came into Sheets as text. Import that tab again with **Convert text to numbers** switched on, then use **Resource › Refresh data** in Looker Studio. |
| A share shows as 0.47% when it should be 47%, or the other way round | Check the field type. `qualifying_share` on **categories** is a fraction and should be **Percent**. `qualifying_share_all_%` on **overall** is already a percentage and should be a plain **Number**. |
| A category number looks far too big | The metric is set to **Sum**. Change it to **Average**: each category appears once in that table, so adding it up multiplies it. |
| The white-space bar chart shows every kind of food | The filter is missing. Add **is_snack Equal to 1** on that chart. Only snack categories have a score. |
| The category filter does not change the bar chart | That is correct. A filter only changes charts built on the same data source. The filter sits on **products**; the bar chart reads **categories**, which already shows every category. |
| A visitor is asked to request access | Set **Data credentials** to **Owner's credentials** on all three sources, and set sharing to **Anyone with the link**. |
| “Invalid column header name(s)… Expected…” | You added a second, different file to one File Upload source. Each of the three tables needs its own data source. |

# 11 What every column means

### dashboard_products.csv — one row per product (a 50,000 sample)

| Column | Meaning |
|---|---|
| code | the barcode; keep it as Text so a leading zero is not lost |
| product_name, brands | the product and who makes it |
| category | the category the notebook worked out |
| is_snack | 1 when the category is one of the seven snack-aisle ones |
| sugars_100g, proteins_100g | grams per 100 g; the two axes of the scatter plot |
| fiber_100g, fat_100g, salt_100g | grams per 100 g; empty where the label did not say |
| energy_kcal_100g | kcal per 100 g |
| health_score | protein minus sugar |
| nutriscore_grade | the A to E grade, where there is one |
| qualifies | 1 when sugar is 5 g or less and protein is 15 g or more |

### category_summary.csv — one row per category, from every cleaned product

| Column | Meaning |
|---|---|
| category | the category name |
| products | how many products it holds |
| qualifying | how many of them meet both limits |
| qualifying_share | that count divided by products; a fraction, so show it as Percent |
| median_sugar, median_protein | the middle value for the category |
| is_snack | 1 for the seven snack-aisle categories |
| white_space_score | the gap score, between 0 and 1; snack categories only |
| white_space_rank | 1 is the biggest gap |

### overall_summary.csv — one row of headline figures

| Column | Meaning |
|---|---|
| rows_read, products_after_cleaning | rows read, and how many survived cleaning |
| snack_products, qualifying_products | products in the snack aisle, and products meeting both limits |
| qualifying_share_all_% | the share of all products that qualify, already multiplied by 100 |
| qualifying_share_snacks_% | the same for the snack aisle only |
| sugar_limit_g, protein_limit_g | the limits used, 5 g and 15 g |

| | |
|---|---|
| `top_category` | the category with the largest white space |
| `target_protein_g`, `target_sugar_g` | the recommended product target |
| `data_last_changed` | when the source file was last updated |